

In much of microeconomic theory, individual preferences are assumed to be *rational*. The hypothesis of rationality is embodied in two basic assumptions about the preference relation $\succsim$: *completeness* and *transitivity*.²

Definition 1.B.1: The preference relation $\succsim$ is *rational* if it possesses the following two properties:

- (i) *Completeness*: for all $x, y \in X$, we have that $x \succsim y$ or $y \succsim x$ (or both).
- (ii) *Transitivity*: For all $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.